

**AI5707 — PATTERN RECOGNITION
AND
MACHINE LEARNING**

ASSIGNMENT – 5

Maximum Likelihood Parameter Estimation

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1. Objective

To perform maximum likelihood parameter estimation for Gaussian class-conditional distributions using both synthetic and real two-dimensional data; estimate the class means and covariance matrices from an 80% training set; classify the remaining 20% test set; visualize Gaussian decision boundaries; compute confusion matrices and classification accuracy; and evaluate the classifiers using ROC curves and AUC.

The synthetic experiment compares isotropic, diagonal and full covariance models. The experiment also studies the effect of equal and different covariance matrices across classes. The real-data experiment applies the same MLE-based Gaussian classification procedure to the supplied two-dimensional dataset.

2. Mathematical Background

2.1. Multivariate Gaussian density

For a d -dimensional feature vector $\mathbf{x}$,

$$p(\mathbf{x} \mid \boldsymbol{\mu}, \Sigma) = \frac{1}{(2\pi)^{d/2} |\Sigma|^{1/2}} \exp \left\{ -\frac{1}{2} (\mathbf{x} - \boldsymbol{\mu})^T \Sigma^{-1} (\mathbf{x} - \boldsymbol{\mu}) \right\}. \quad (1)$$

2.2. Maximum likelihood estimates

For N samples belonging to one class, the MLE of the mean is

$$\hat{\boldsymbol{\mu}} = \frac{1}{N} \sum_{n=1}^N \mathbf{x}_n. \quad (2)$$

The MLE covariance is

$$\hat{\Sigma} = \frac{1}{N} \sum_{n=1}^N (\mathbf{x}_n - \hat{\boldsymbol{\mu}})(\mathbf{x}_n - \hat{\boldsymbol{\mu}})^T. \quad (3)$$

The denominator is N , since this is maximum likelihood estimation rather than the unbiased sample covariance estimator.

2.3. Covariance models

For the isotropic model,

$$\Sigma = \sigma^2 I. \quad (4)$$

For the diagonal model,

$$\Sigma = \begin{bmatrix} \sigma_1^2 & 0 \\ 0 & \sigma_2^2 \end{bmatrix}. \quad (5)$$

For the full covariance model,

$$\Sigma = \begin{bmatrix} \sigma_{11} & \sigma_{12} \\ \sigma_{21} & \sigma_{22} \end{bmatrix}. \quad (6)$$

2.4. Gaussian discriminant

For class k , assuming equal class priors,

$$g_k(\mathbf{x}) = -\frac{1}{2} \ln |\Sigma_k| - \frac{1}{2} (\mathbf{x} - \boldsymbol{\mu}_k)^T \Sigma_k^{-1} (\mathbf{x} - \boldsymbol{\mu}_k). \quad (7)$$

The predicted class is

$$\hat{k} = \arg \max_k g_k(\mathbf{x}). \quad (8)$$

The decision boundary between classes i and j satisfies

$$g_i(\mathbf{x}) = g_j(\mathbf{x}). \quad (9)$$

Equal covariance matrices cancel the quadratic terms and therefore give a linear boundary. Different covariance matrices generally give a quadratic boundary.

2.5. ROC measures

For one-vs-rest ROC analysis,

$$\text{TPR} = \frac{TP}{TP + FN}, \quad \text{FPR} = \frac{FP}{FP + TN}. \quad (10)$$

The ROC curve plots TPR against FPR as the classification threshold is varied. The area under this curve is the AUC.

3. Experimental Parameters

Three Gaussian classes were generated with the following class means:

$$\boldsymbol{\mu}_1 = [2, 2]^T, \quad \boldsymbol{\mu}_2 = [7, 3.5]^T, \quad \boldsymbol{\mu}_3 = [4, 8]^T. \quad (11)$$

The class-specific full covariance matrices used to generate the synthetic data were

$$\Sigma_1 = \begin{bmatrix} 1.5 & 0.4 \\ 0.4 & 1.0 \end{bmatrix}, \quad \Sigma_2 = \begin{bmatrix} 1.0 & -0.3 \\ -0.3 & 1.8 \end{bmatrix}, \quad \Sigma_3 = \begin{bmatrix} 1.8 & 0.5 \\ 0.5 & 1.2 \end{bmatrix}.$$

A total of 10,000 samples were generated for each class, giving 30,000 synthetic samples. An 80:20 stratified split gives 24,000 training samples and 6,000 test samples.

The real-data files were supplied in the format (x, y, class) . Combining the two supplied files gives 1,350 observations, followed by an 80:20 stratified split into 1,080 training and 270 test samples.

4. Implementation Procedure

- Generate the three synthetic Gaussian classes using a fixed random seed.
- Split the synthetic data into 80% training and 20% testing subsets.
- Estimate the class mean and covariance using MLE.
- Evaluate isotropic, diagonal and full covariance models.
- Compute Gaussian discriminant scores for every test sample.
- Assign each test point to the class with the largest discriminant score.
- Plot the estimated decision boundaries and test samples.
- Compute confusion matrices and classification accuracy.
- Plot one-vs-rest ROC curves and calculate AUC.
- Repeat the analysis for shared and class-specific covariance matrices.
- Apply the full-covariance MLE classifier to the supplied real dataset.

5. Synthetic Data Experiment

5.1. Generated synthetic data

Figure 1 shows the generated three-class synthetic dataset. The three Gaussian clusters are well separated, although some overlap is present near the class interfaces.

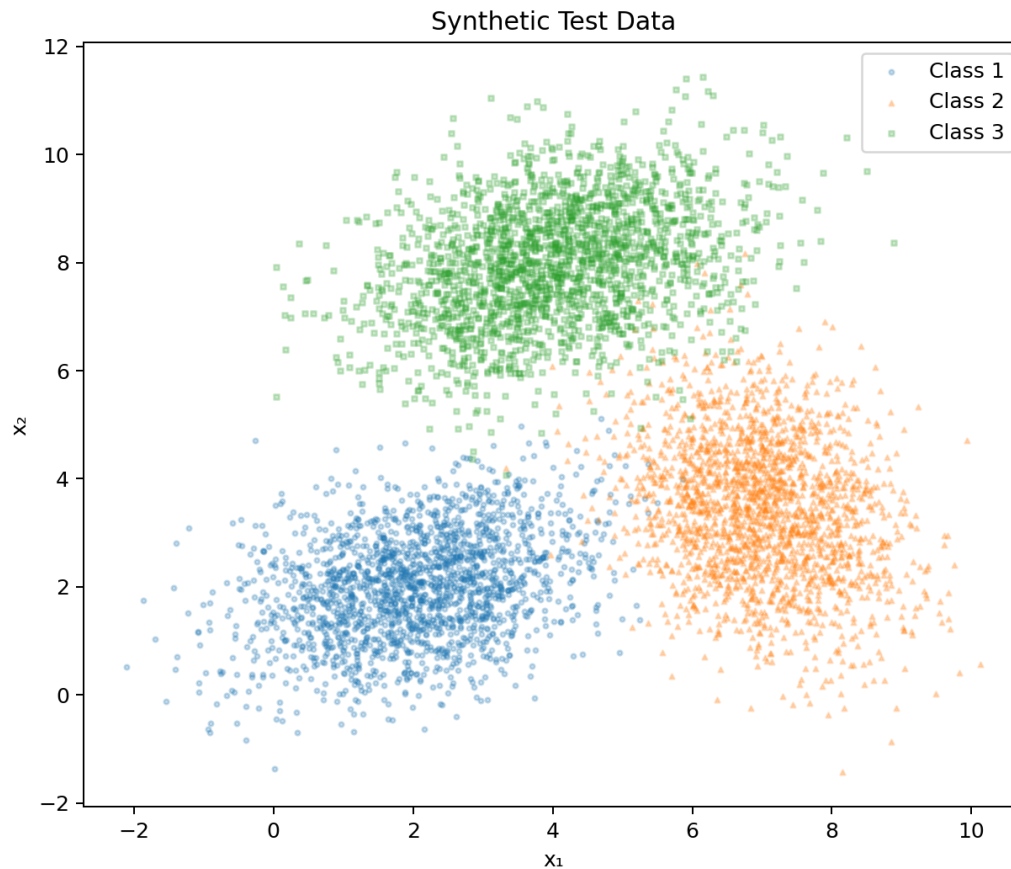


Figure 1: Synthetic Gaussian data for the three classes.

5.2. Classification results

The test-set accuracies obtained using the three covariance models are shown below.

Table 1: Synthetic-data classification accuracy

Covariance model	Accuracy
Isotropic ($\sigma^2 I$)	97.95%
Diagonal	98.52%
Full	98.87%

5.3. Confusion matrices

For the isotropic model, the confusion matrix is

$$\begin{bmatrix} 1940 & 54 & 6 \\ 3 & 1952 & 45 \\ 13 & 2 & 1985 \end{bmatrix}.$$

For the diagonal model, the confusion matrix is

$$\begin{bmatrix} 1967 & 27 & 6 \\ 8 & 1958 & 34 \\ 10 & 4 & 1986 \end{bmatrix}.$$

For the full covariance model, the confusion matrix is

$$\begin{bmatrix} 1979 & 15 & 6 \\ 14 & 1971 & 15 \\ 7 & 11 & 1982 \end{bmatrix}.$$

5.4. Decision boundaries

Figure 2 shows the Gaussian decision regions and boundaries obtained from the estimated class parameters.

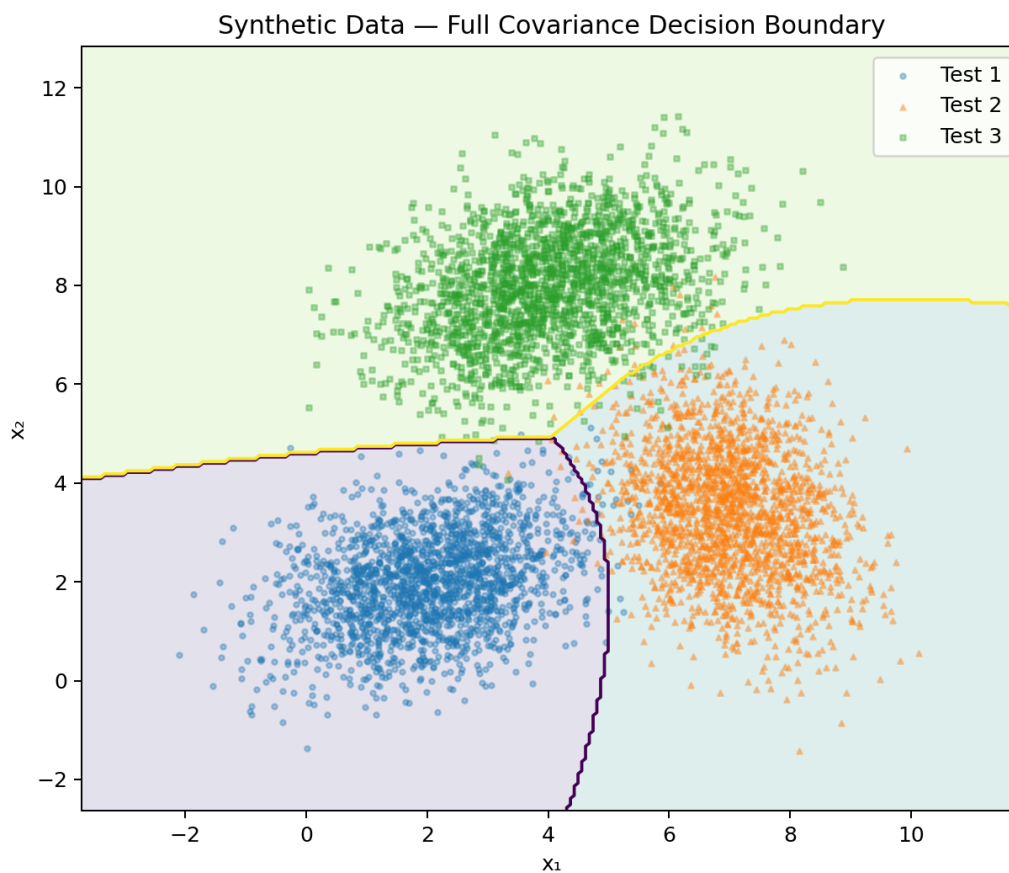


Figure 2: Gaussian decision boundaries for the synthetic test data.

6. ROC Analysis

6.1. Isotropic covariance

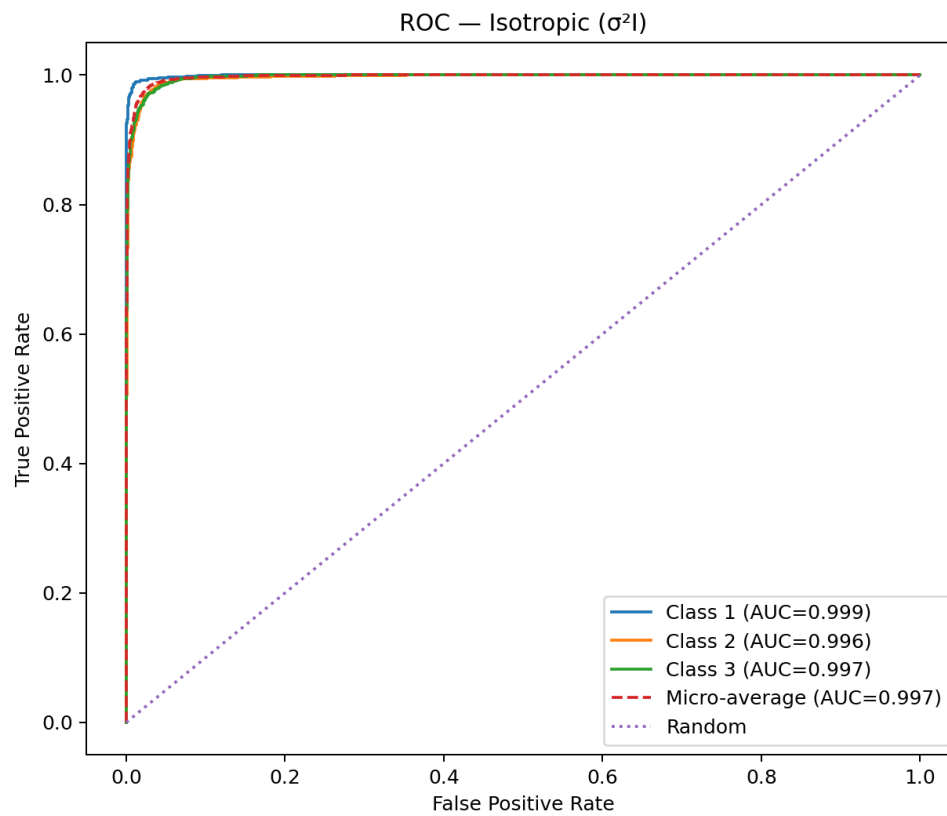


Figure 3: One-vs-rest ROC curves for the isotropic covariance model.

6.2. Diagonal covariance

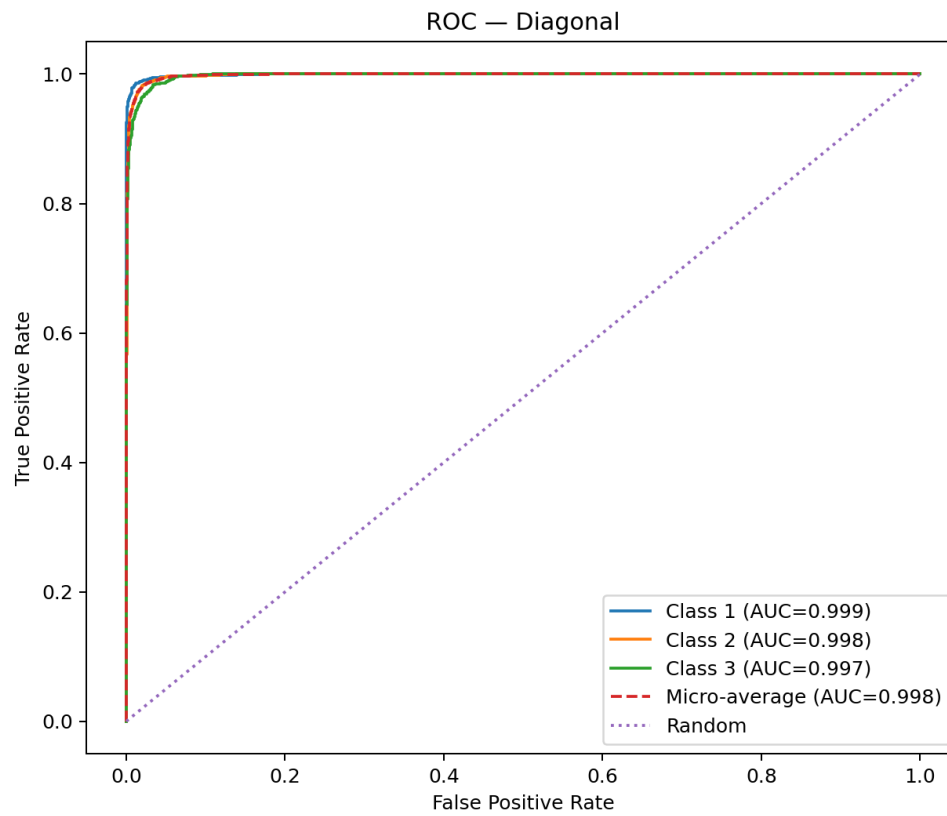


Figure 4: One-vs-rest ROC curves for the diagonal covariance model.

6.3. Full covariance

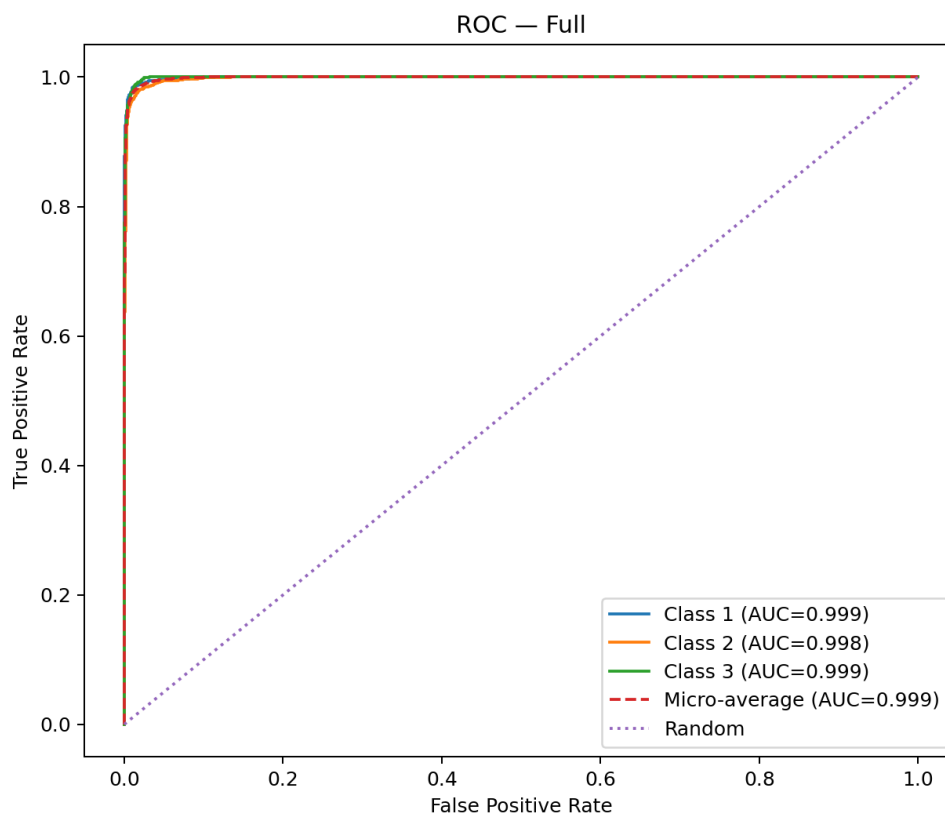


Figure 5: One-vs-rest ROC curves for the full covariance model.

The corresponding AUC values are

Table 2: Synthetic-data ROC-AUC results

Model	Class 1	Class 2	Class 3	Micro-average
Isotropic	0.9992	0.9960	0.9967	0.9974
Diagonal	0.9991	0.9979	0.9973	0.9981
Full	0.9990	0.9979	0.9991	0.9987

7. Equal and Different Covariance Matrices

When the covariance matrix is shared by all classes, the Gaussian discriminants have the same quadratic term. These terms cancel when two discriminants are subtracted, producing a linear decision boundary.

For class-specific covariance matrices, the quadratic terms do not cancel. Consequently, the decision boundary is generally quadratic.

The synthetic experiment demonstrates that allowing a separate covariance matrix for each class gives the classifier more freedom to model class orientation, spread and correlation.

8. Real 2-D Data Experiment

The supplied real data contain two numerical features and a class label. The combined dataset contains 1,350 observations. A stratified 80:20 split gives 1,080 training samples and 270 test samples.

A full covariance Gaussian model was estimated from the training set.

8.1. Estimated class means

Table 3: MLE estimates of class means

Class	$\hat{\mu}_1$	$\hat{\mu}_2$
1	425.5762	1231.8512
2	342.5047	2026.6825
3	576.1742	1446.4665

8.2. Estimated covariance matrices

$$\hat{\Sigma}_1 = \begin{bmatrix} 28772.5232 & 34029.2726 \\ 34029.2726 & 163995.7691 \end{bmatrix},$$

$$\hat{\Sigma}_2 = \begin{bmatrix} 6345.6194 & -5984.9171 \\ -5984.9171 & 44911.2623 \end{bmatrix},$$

$$\hat{\Sigma}_3 = \begin{bmatrix} 24779.3116 & 6193.8729 \\ 6193.8729 & 63158.5250 \end{bmatrix}.$$

8.3. Classification performance

The real-data test accuracy is

$$\boxed{85.93\%}. \quad (12)$$

The confusion matrix is

$$\begin{bmatrix} 68 & 2 & 20 \\ 3 & 85 & 2 \\ 7 & 4 & 79 \end{bmatrix}.$$

8.4. Decision boundary

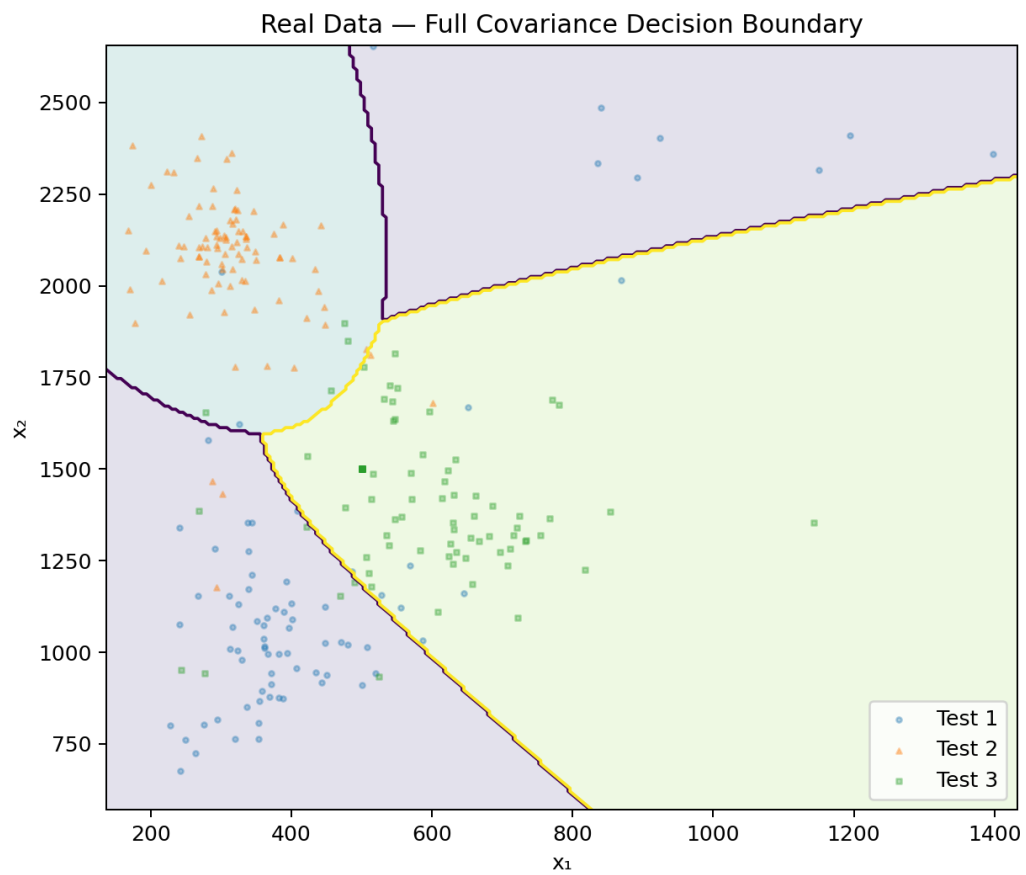


Figure 6: Gaussian decision boundaries for the real two-dimensional dataset.

8.5. ROC curve

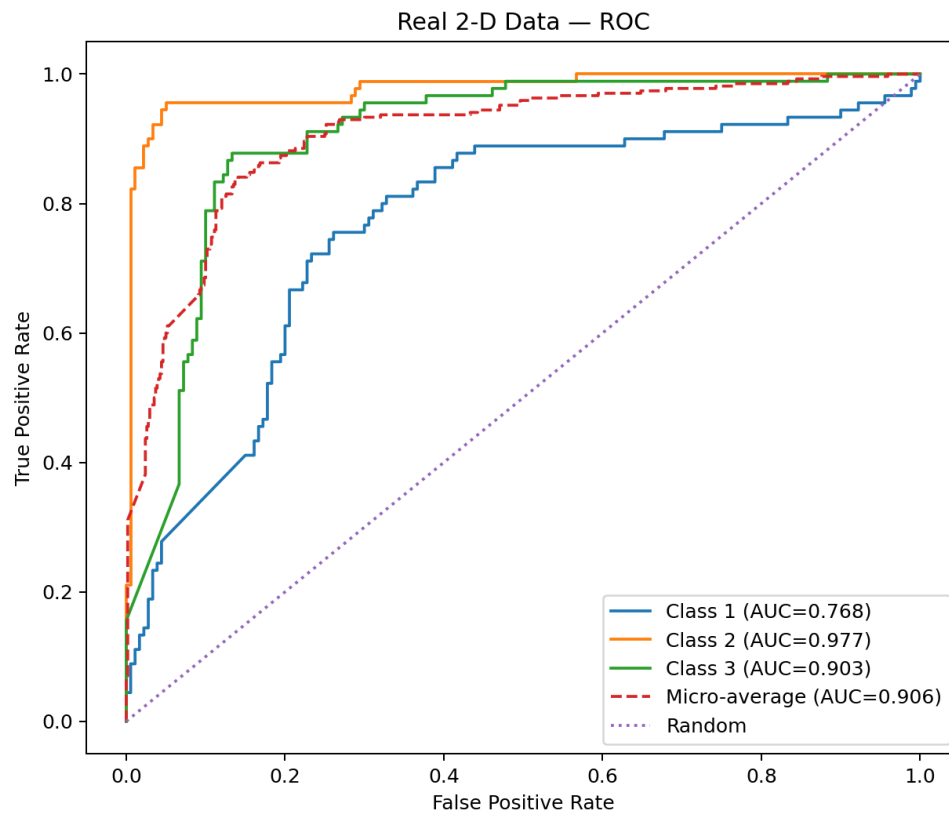


Figure 7: One-vs-rest ROC curves for the real dataset.

The real-data AUC values are

Table 4: Real-data ROC-AUC results

Measure	AUC
Class 1	0.7681
Class 2	0.9770
Class 3	0.9030
Micro-average	0.9057

9. Results and Discussion

- The isotropic model achieved 97.95% accuracy. Its restricted covariance structure cannot represent different spreads or correlations independently.
- The diagonal model improved the accuracy to 98.52% because it allows different variances along the two feature dimensions.
- The full covariance model achieved the best synthetic accuracy, 98.87%, because it additionally captures feature correlation.

- The synthetic ROC-AUC values are very close to one, confirming strong separability between the generated classes.
- Equal covariance matrices lead to linear Gaussian decision boundaries, whereas different covariance matrices generally lead to curved quadratic boundaries.
- The real dataset is more challenging than the synthetic dataset, with a test accuracy of 85.93%.
- For the real data, Class 2 has the strongest one-vs-rest separation with an AUC of approximately 0.9770, while Class 1 is comparatively harder with an AUC of approximately 0.7681.

10. Observations

The experiment demonstrates that covariance structure has a direct effect on the Gaussian classifier. Increasing the flexibility from isotropic to diagonal and then to full covariance generally improves the ability of the model to represent the actual class distributions.

The decision-boundary plots provide a geometric interpretation of the discriminant functions. The ROC plots provide a threshold-independent view of classification quality, complementing the accuracy and confusion-matrix results.

The real-data experiment also demonstrates the practical effect of overlap between classes: even with a full covariance model, the classification accuracy is lower than for the deliberately well-separated synthetic Gaussian classes.

11. Conclusion

The assignment successfully demonstrates maximum likelihood parameter estimation for Gaussian class-conditional models. The estimated means and covariance matrices were used for classification through Gaussian discriminant functions.

For the synthetic dataset, the full covariance model achieved the highest test accuracy of 98.87% and a micro-average ROC-AUC of 0.9987. The diagonal model performed slightly better than the isotropic model because it can represent different variances along the two feature axes.

For the supplied real two-dimensional data, the full covariance model achieved 85.93% test accuracy with a micro-average ROC-AUC of approximately 0.9057.

Overall, the experiment shows how MLE, covariance structure, Gaussian discriminants, decision boundaries, confusion matrices and ROC analysis work together in a pattern-recognition classification pipeline.

12. Software and Tools

- Python
- NumPy

- Pandas
- Matplotlib
- Scikit-learn
- Jupyter Notebook
- LaTeX / Overleaf